Using ADALM-Pluto for Monotone and Audio Signal Transmission/Reception

Department of Electrical and Computer Engineering
Birzeit University
Communication Systems, ENEE3309

Basmala abu hakema [1], *Yasmin Al shawawrh* [2], *Layal hajji* [3]

1220184@student.birzeit.edu[1], 1220848@student.birzeit.edu[2], 1220871@student.birzeit.edu[3]

Abstract

This project focuses on creating a communication system using the ADALM-Pluto SDR (Software Defined Radio) and GNU Radio software. The goal is to design a system that can send and receive both simple tone signals (like sine waves) and audio signals using Normal AM modulation. The process involves generating tone signals, modulating them, and transmitting them. Then, the signals are received, demodulated, and checked to see if the original tone is recovered clearly. For audio signals, the system captures and sends the audio, and the received audio is tested for clarity. To improve the system, adjustments are made to factors like power, receiver settings, and filters to reduce noise and improve signal quality. Tools like QT GUI and spectral analysis are used to visually check the performance of the system. This project helps understand key concepts of software-defined radio, signal processing, and how to optimize modern communication systems.

Index Terms - ADALM-Pluto, communication systems, GNU Radio, Normal AM, signal processing, software-defined radio.

1. Introduction

Amplitude Modulation (AM) is a method used to send information, like sound or data, over radio waves. In AM, the strength (or amplitude) of a carrier wave is changed based on the information signal. For example, when transmitting audio, the amplitude of the carrier wave adjusts to match the sound's vibrations.

AM is one of the earliest and simplest types of modulation used in communication systems. It is commonly used for broadcasting on AM radio and in aviation communication. Although newer methods like FM and digital signals have become more popular, AM is still widely used due to its simplicity and effectiveness over long distances.

2. System Design, Experiments and Results

In general, the transmission and receiver flow graphs work together to implement a complete communication system. The transmitter generates and modulates signals, processes them for SDR compatibility, and transmits them over the air using the ADALM-Pluto SDR. The receiver captures these signals, demodulates them to recover the original information, and outputs the results for evaluation. This design accommodates both monotone and audio signals, providing hands-on experience with real-world communication system design.

2.1. Monotone signal

2.1.1. Monotone Signal Transmission

The transmission system is designed to generate and transmit both monotone and audio signals using amplitude modulation (AM). For monotone signals, the flow graph includes an **Analog Signal Source** producing a sine wave with a configurable frequency and amplitude, serving as the information signal. This signal is first combined with a constant value from a **Constant Source** using a **Multiply Block**, ensuring the proper amplitude for modulation. The output is further processed with another **Constant Source** and combined using an **Add Block**.

The modulated signal is created by multiplying the processed information signal with a carrier wave, generated by another **Signal Source (Carrier Wave Source)**, using a second **Multiply Block**. To prepare the signal for transmission, the modulated signal is converted into a complex format using a **Float to Complex Converter**, which is necessary for compatibility with the ADALM-Pluto SDR. The complex signal is sent to the **Pluto Sink**, transmitting the signal over the air. For real-time monitoring, a **QT GUI Sink** is included to visualize the transmitted signal's frequency spectrum and time-domain waveform.

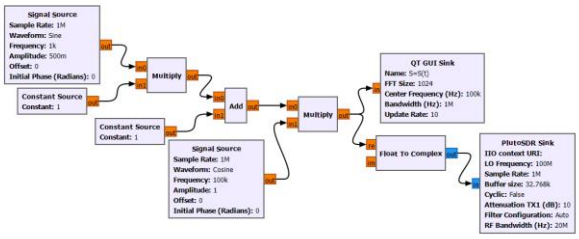

Figure 1: *monotone transmission*

2.1.2. *Monotone Signal Reception*

The monotone reception system is designed to capture and process amplitude-modulated (AM) signals to recover the original monotone information signal. It begins with a **Pluto Source**, which receives the modulated signal over the air. The received complex signal is converted to its magnitude using a **Complex to Mag** block, enabling envelope detection.

The detected signal passes through two **Low Pass Filters**, which remove high-frequency noise and isolate the baseband monotone signal. An **Absolute Value** block is also included in the processing chain to refine the detected envelope. To further enhance the signal, a **DC Blocker** eliminates any DC offset, ensuring a clean signal.

For real-time monitoring and analysis, the system includes multiple visualization tools such as **QT GUI Time Sinks** and **QT GUI Frequency Sinks**, which display the time-domain waveform and frequency spectrum of both the received and recovered signals. This setup ensures accurate reception and reconstruction of the transmitted monotone signal.

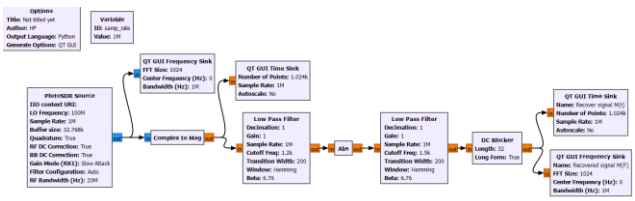

Figure 2: *monotone reception*

2.1.3. *Monotone transmission reception system*

In this part, we designed a combined transmitter and receiver system using GNU Radio to validate the modulation and demodulation processes without relying on external hardware like PlutoSDR. The setup connects a sine wave signal source, representing the input message signal, with modulation and transmission blocks. These signals are then combined, processed, and directly fed into the demodulation and receiver path for recovery.

The graph demonstrates the flow, starting with the generation of two orthogonal signal sources (sine and cosine waves). These signals are modulated using multipliers and summed together to simulate modulation. The resulting modulated signal is then converted to a complex signal, passed through a magnitude computation, and filtered using two low-pass filters to extract the baseband components.

Additional processing includes absolute value computation and DC blocking to improve signal clarity. The recovered signal is finally analyzed using time and frequency domain QT GUI sinks. This approach ensures the transmitted and received signals can be observed and compared in real-time, confirming accurate modulation and demodulation functionality.

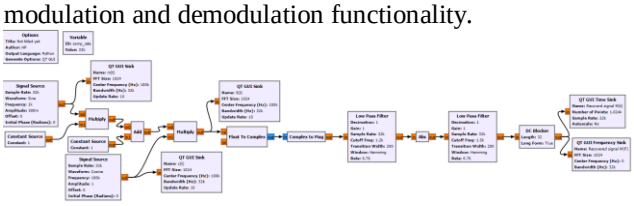

Figure 3: *monotone transmission reception system*

Testing Monotone Modulation and Demodulation Using GNU Radio Companion Without Hardware:

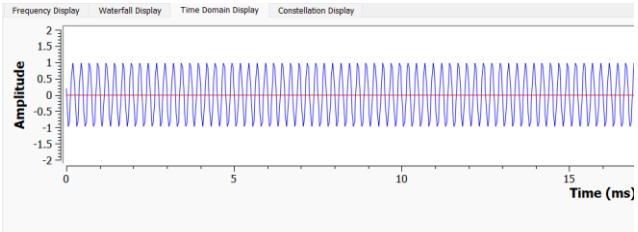

Figure 4: *c(t)*

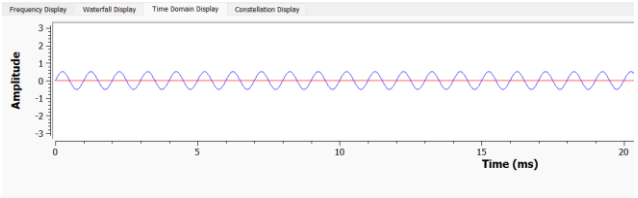

Figure 5: *m(t)*

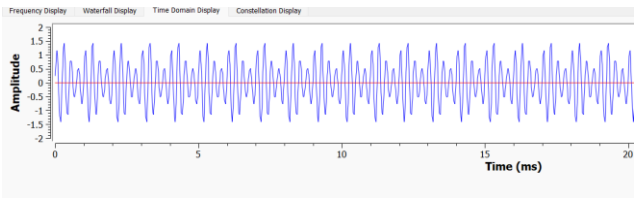

Figure 6: s(t)

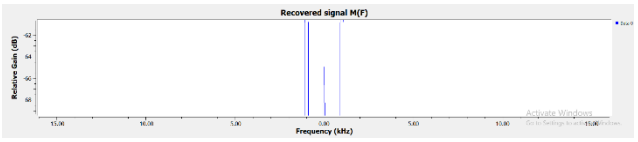

Figure 7: *m(F) recovered*

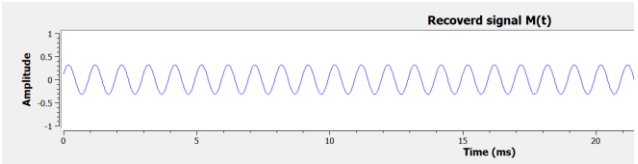

Figure 8: *m(t) recovered*

2.2. Audio signal

2.2.1. Audio Signal Transmission

In this part, we transmitted an audio signal modulated onto a carrier using the PlutoSDR. The WAV File Source block reads an audio file ("Rhymes For Kids.wav") repeatedly and outputs the audio samples. These samples are processed through a Rational Resampler, which interpolates and decimates the signal to achieve the desired sampling rate while maintaining quality using specified fractional bandwidth and taps. A Constant Source adds a DC offset to the audio signal. The resulting signal is then Multiplied with a cosine wave generated by the Signal Source, which has a frequency of 300 kHz, creating an amplitude-modulated (AM) signal.

This modulated signal is further combined with a Constant Source set to 1 for normalization or scaling. A second Multiply operation combines the resulting signal with another signal if necessary. The processed signal is converted into a complex format using the Float To Complex block, which separates real and imaginary components for further processing.

The QT GUI Sink provides a visual representation of the signal's spectrum, showing the modulated signal in the frequency domain. Finally, the PlutoSDR Sink transmits the modulated signal via the ADALM-PlutoSDR hardware at a configured LO frequency of 2.4 GHz, with adjustable RF bandwidth, attenuation, and buffer size for optimal transmission. This setup effectively modulates and transmits the audio signal.

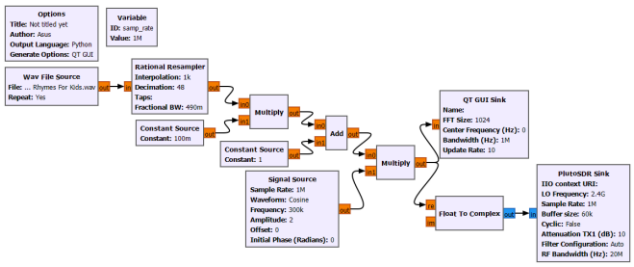

Figure 9: *Audio transmission*

2.2.2. Audio Signal Reception

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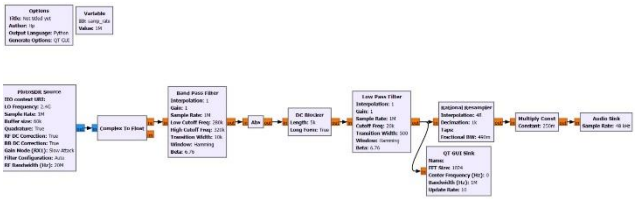

Figure 10: *Audio reception*

2.2.3 Audio transmission reception system

This flowgraph represents a software-defined radio (SDR) signal processing chain implemented in GNU Radio. It begins with a PlutoSDR Source block, configured to operate at a center frequency of 2.4 GHz with a sampling rate of 1 MS/s and an RF bandwidth of 20 MHz. The output of the source is a complex signal processed by a Complex to Float block, splitting the complex signal into its real and imaginary components.

The signal is filtered using a Band Pass Filter, with a sample rate of 1 MS/s, low and high cutoff frequencies of 280 kHz and 320 kHz, respectively, a transition width of 10 kH. The absolute value of the signal is then taken with an Abs block, followed by a DC Blocker with a length of 5000 samples, configured in long-form mode to remove any DC component.

The filtered signal is further processed by a Low Pass Filter, designed with a cutoff frequency of 20 kHz, a transition width of 500 Hz. A Rational Resampler is applied next, with an interpolation factor of 48 and a decimation factor of 1000, reducing the sampling rate for audio output while maintaining signal integrity.

The signal's amplitude is scaled using a Multiply Const block, where the multiplication factor is set to 250m. Finally, the processed signal is sent to two sinks: a QT GUI Sink for visualizing the signal's frequency spectrum and an Audio Sink with a sample rate of 48 kHz for audio playback. The QT GUI Sink is configured with an FFT size of 1024, a bandwidth of 1 MHz, and a center frequency of 0 Hz, updating every 10 ms.

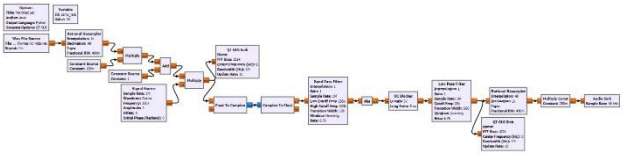

Figure 11*: Audio transmission reception system*

In our system without the PlutoSDR device, we received the audio but the recovered audio signal exhibited noticeable noise due to the inherent characteristics of Normal Amplitude Modulation (AM). Normal AM has a relatively low efficiency of 33.3%, as only one-third of the transmitted power carries the actual message, while the remaining power is distributed between the carrier and redundant sideband. Additionally, Normal AM has a bandwidth of 2fm where fm is the highest frequency in the modulating signal. This wide bandwidth increases susceptibility to noise and interference, degrading the quality of the recovered signal. These factors combined make Normal AM less efficient and more prone to noise, impacting the clarity of the audio output in our test.

Testing Audio Modulation and Demodulation Using GNU Radio Companion Without Hardware:

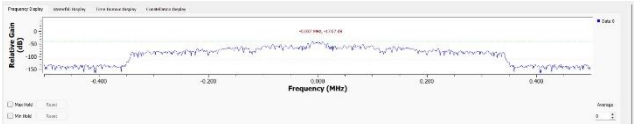

Figure 12: *Audio message signal m(t)*

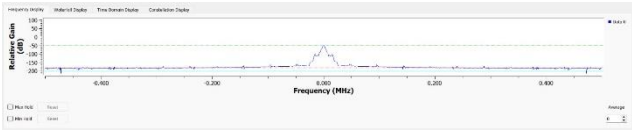

Figure 13*: After the first multiply*

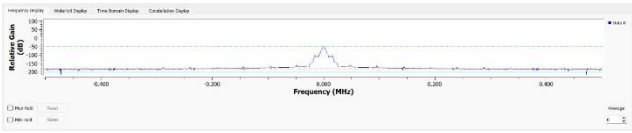

Figure 14*: After the add*

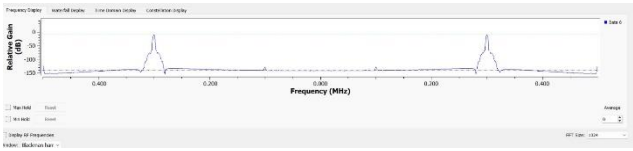

Figure 15*: After the second multiply*

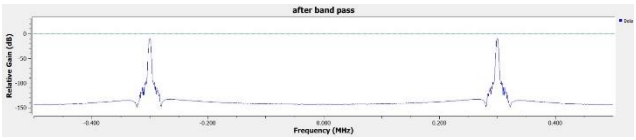

Figure 16*: After the Band pass*

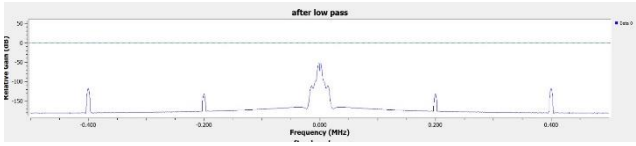

Figure 17*: After the low pass*

The recovered signal appears weaker and less well-tuned due to the inherent inefficiency of AM modulation, which operates at only 33% efficiency. Additionally, the bandwidth (BW) of the signal is twice the frequency of the modulating signal (2fm), further contributing to the reduced quality of the recovered signal. This combination of factors results in a weaker, less accurate reproduction of the original signal.

2.3 Practical Testing: Audio Signal Transmission/Reception Using ADALM-Pluto SDR with GNU Radio

We tested the transmitter and receiver system using the ADALM-Pluto SDR, ensuring proper operation and signal flow through the implemented blocks. Below, we analyzed the outputs at various stages in the transmitter using QT GUI Sink visualizations. These include the message signal after the QT GUI sink, the output after the first multiplication block, the signal after the addition block, and finally, the signal post the second multiplication block. Each stage was verified for accuracy and alignment with expected signal behavior.

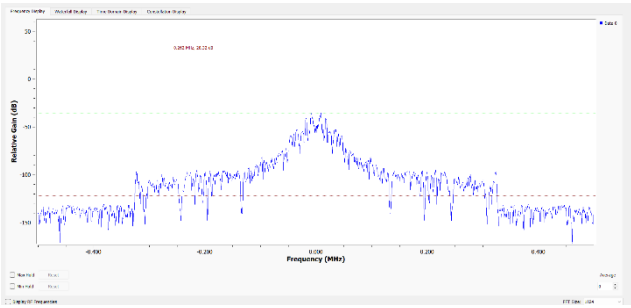

Figure 18: Audio message signal m(t)

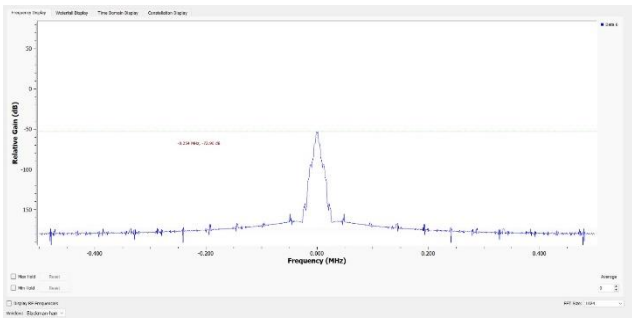

Figure 19*: After the first multiply*

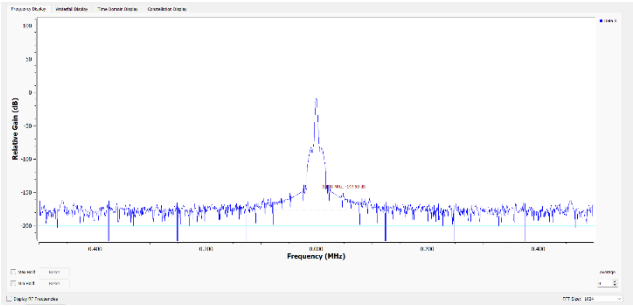

Figure 20*: After the add*

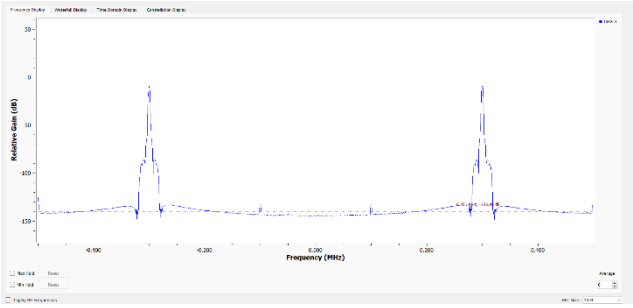

Figure 21*: After the second multiply*

We tested the receiver system using the ADALM-Pluto SDR to ensure accurate signal processing and recovery. Below, we analyzed the outputs at various stages in the receiver using QT GUI Sink visualizations. These include the signal after the Band Pass Filter, where the desired frequency components are isolated, and the signal after the Low Pass Filter, where noise is minimized to recover the transmitted signal. Each stage was carefully verified for correctness and performance.

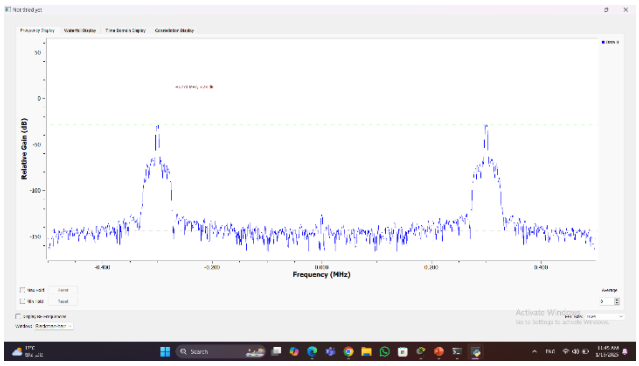

Figure 22*: After the Band pass filter*

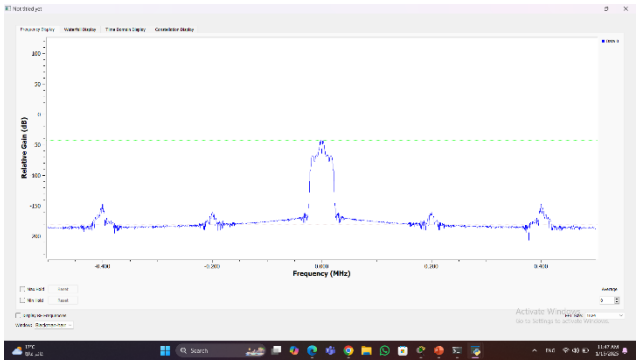

Figure 23*: After the low pass filter*

2.4 Parameter Tuning

Note: The values in the audio transmission (figure 9) and audio reception (figure 10) figures represent the parameters that achieved the most clarity after practical testing with the Pluto SDR.

The cutoff frequency in the low-pass filter is a critical parameter in ensuring effective signal demodulation and clear audio output. For the receiver in this AM system, the low-pass filter was set with a cutoff frequency of 20 kHz, which aligns with the typical audio signal range of 10–20 kHz. This value was chosen to allow the full audio spectrum to pass while filtering out higher-frequency noise. A properly selected cutoff frequency is essential to preserve signal fidelity, as a lower cutoff might attenuate important parts of the audio signal, and a higher cutoff would allow excessive noise to pass through. Additionally, the transition width of the filter was optimized to provide a smooth roll-off, balancing noise reduction with the preservation of audio quality.

The rational resampler, a critical block in the signal processing chain, is used to adjust sample rates for compatibility with different system components. On the transmitter side, the original sample rate of 1 MHz was maintained, matching the sample rate required by the PlutoSDR Sink for transmission. On the receiver side, the rational resampler decimated the signal by a factor of 1,000 and interpolated it by a factor of 48, effectively bringing the sample rate down to 48 kHz. This adjustment was crucial to align the signal with the audio sink's requirements, enabling accurate audio playback. Proper resampling ensures that the Nyquist criteria are satisfied, preventing aliasing and maintaining signal fidelity throughout the system.

The band-pass filter in the receiver, with a low cutoff frequency of 280 kHz(fc-fm)and a high cutoff of 320 kHz (fc+fm), was designed to isolate the AM carrier frequency and its immediate sidebands. This narrow passband was crucial for rejecting adjacent channel interference and noise. However, despite the careful tuning of the filters and other parameters, some noise persisted in the system, which is a characteristic limitation of normal AM modulation. AM's inherent efficiency of just 33% means that a significant portion of the transmitted power is allocated to the carrier, leaving only a small fraction for the information-carrying sidebands. This inefficiency, coupled with AM's relatively large bandwidth, contributes to the presence of noise, even after extensive optimization.

Additionally, the bandwidth of AM signals adds to the complexity of achieving clean reception. Even with adjustments to parameters like cutoff frequencies, resampling ratios, and receiver gain, some degree of noise remains unavoidable. This is further exacerbated by environmental interference and AM's susceptibility to

noise. These limitations highlight the trade-offs in AM systems: simplicity and compatibility with existing infrastructure are achieved at the expense of efficiency and noise resilience. Ultimately, the system was optimized to provide acceptable audio quality, but the presence of noise reflects the inherent challenges of AM communication.

Conclusions

This project successfully demonstrated the transmission and reception of monotone and audio signals using ADALM-Pluto SDR and GNU Radio with normal amplitude modulation (AM). The experiments validated the system's ability to effectively generate, transmit, and recover signals while maintaining high spectral and signal fidelity. Despite some noise present, parameter optimization played a critical role in minimizing noise, improving signal clarity, and ensuring robust system performance under various conditions. Specifically, adjustments to transmit power, receiver gain, and low-pass filter characteristics were pivotal in achieving optimal results.

Future improvements could involve exploring advanced modulation schemes like frequency modulation (FM) or digital modulation techniques such as QAM to enhance the system's versatility. Additionally, implementing adaptive filtering techniques could further improve signal quality in noisy environments. Extending the system to support multi-channel or multi-user scenarios could provide insights into handling more complex communication tasks. This project underscores the potential of SDR platforms in enabling flexible and efficient communication system designs for both educational and professional applications.

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   [Accessed on 10/1/2025 8:47pm]

3. https://en.wikipedia.org/wiki/Amplitude_modulation [Accessed on 10/1/2025 9:25pm]

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